



Inspiring Excellence

Assignment-03

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Course: CSE330

Section: 13

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Ans. to the ques. no-01

a

$$f(x) = 2x^3 - 3x^2 - 11x + 6$$

$$2x^3 - 3x^2 - 11x + 6 = 0$$

$$\text{or, } 2x^3 - x^2 - 2x^2 + x - 12x + 6 = 0$$

$$\text{or, } x^2(2x-1) - x(2x-1) + 6(2x-1) = 0$$

$$\text{or, } (2x-1)(x^2 - x - 6) = 0$$

$$\text{or, } (2x-1)(x^2 - 3x + 2x - 6) = 0$$

$$\text{or, } (2x-1)\{x(x-3) + 2(x-3)\} = 0$$

$$\text{or, } (2x-1)\{(x-3)(x+2)\} = 0$$

$$\therefore (2x-1)(x-3)(x+2) = 0$$

$$\therefore x = \frac{1}{2}, 3, -2$$

$\therefore$ The other two roots are: 3, -2

6

$$f(x) = 2x^3 - 3x^2 - 11x + 6$$

$$2x^3 - 3x^2 - 11x + 6 = 0$$

$$\text{or, } 11x = 2x^3 - 3x^2 + 6$$

$$\therefore x = \frac{2x^3 - 3x^2 + 6}{11}$$

$$\therefore g_1(x) = \frac{2x^3 - 3x^2 + 6}{11}$$

$$2x^3 - 3x^2 - 11x + 6 = 0$$

$$\text{or, } 3x^2 = 2x^3 - 11x + 6$$

$$\text{or, } x^2 = \frac{2x^3 - 11x + 6}{3}$$

$$\therefore x = \sqrt{\frac{2x^3 - 11x + 6}{3}}$$

$$\therefore g_2(x) = \sqrt{\frac{2x^3 - 11x + 6}{3}}$$

$$g_1(x) = \frac{2x^3 - 3x^2 + 6}{11}$$

$$\lambda = |g'_1(x)| = \left| \frac{6x^2 - 6x}{11} \right|$$

$$\left| \frac{6x^2 - 6x}{11} \right|$$

$$x_* = 0.5$$

$$|g'_1(0.5)| = 0.137 < 1$$

Convergence

$$x_* = 3$$

$$|g'_1(3)| = 3.27 > 1$$

Divergence

$$x_* = -2$$

$$|g'_1(-2)| = 3.27 > 1$$

Divergence

$$g_2(x) = \sqrt{\frac{2x^3 - 11x + 6}{3}}$$

$$\lambda = |g'_2(x)| = \left| \frac{1}{6} \cdot \frac{1}{\sqrt{\frac{2x^3 - 11x + 6}{3}}} \cdot (6x^2 - 11) \right|$$

$$= \left| \frac{1}{6} \cdot \frac{6x^2 - 11}{\sqrt{\frac{2x^3 - 11x + 6}{3}}} \right|$$

$$x_* = 0.5$$

$$|g'_2(0.5)| = 3.17 > 1$$

Divergence

$$x_* = 3$$

$$|g'_2(3)| = 2.39 > 1$$

Divergence

$$x_* = -2$$

$$|g'_2(-2)| = 1.08 > 1$$

Divergence